

PROJECT PROPOSAL

IT Asset & Hardware Maintenance Log System

A Database Management System (DBMS) Project Proposal

Submitted By: MD Ahsanul Kabir Mohit - 2241091304
Sajedul hasan - 2241091323
Md Arman Khan - 2241091341
Himel Biswas - 2241091345



Department: Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE)

Project Type: Database Management System (DBMS) Project

Duration: 6 – 8 Weeks

Version: 1.0

Submission Date: July 31, 2026

Submitted in partial fulfilment of the requirements of the Database Management Systems course

Table of Contents

1. Introduction	3
2. Problem Statement.....	3
3. Proposed Solution	3
4. Project Objectives	4
5. Scope of the Project.....	4
6. Functional Requirements.....	5
7. Non-Functional Requirements.....	6
8. System Features	6
9. System Modules.....	7
10. Database Design.....	8
11. Entity Relationship Diagram (ERD).....	9
12. Data Flow Diagram (Level 0)	10
13. Technology Stack	10
14. User Interface Overview	10
15. Testing Plan	11
16. Expected Outcomes	12
17. Future Enhancements	12
18. Conclusion.....	12
19. References	13

1. Introduction

Managing IT assets is an essential responsibility for every organization. Computers, laptops, printers, routers, monitors, and networking equipment must be properly tracked to ensure efficient utilization and timely maintenance.

Many small businesses and educational institutions still rely on spreadsheets or paper records to manage their IT equipment. This often leads to missing devices, inaccurate records, delayed maintenance, and poor asset utilization.

The IT Asset & Hardware Maintenance Log System is a simple database-driven application that allows organizations to maintain records of all IT assets, assign equipment to users or departments, monitor maintenance schedules, and update equipment status in real time.

2. Problem Statement

Small IT offices often experience difficulties because:

- Equipment records are maintained manually.
- Asset ownership is difficult to track.
- Maintenance history is not properly recorded.
- Devices needing repair are forgotten.
- Reports take considerable time to prepare.
- Duplicate asset records are common.
- Missing assets cannot be identified quickly.

These issues reduce productivity and increase maintenance costs.

3. Proposed Solution

The proposed system provides a centralized database that stores information about every IT asset. The system enables administrators to:

- Register new hardware assets.
- Assign equipment to employees or departments.
- Track current asset status.
- Record maintenance activities.
- Search assets instantly.
- Generate maintenance reports.

4. Project Objectives

The main objectives of this project are:

1. Develop a centralized IT asset database.
2. Track all hardware assets efficiently.
3. Reduce paperwork.
4. Improve maintenance scheduling.
5. Increase equipment accountability.
6. Minimize asset loss.
7. Simplify reporting.
8. Improve overall IT resource management.

5. Scope of the Project

5.1 In Scope

The project covers the following categories of hardware:

- Desktop computers
- Laptops
- Printers
- Routers
- Switches
- Monitors
- UPS devices
- Projectors
- Network devices
- Other IT hardware

5.2 Out of Scope

The following are explicitly excluded from the current version of the project:

- Software license management
- Automatic hardware detection
- Barcode scanning
- Mobile application
- Cloud synchronization

6. Functional Requirements

6.1 Asset Management

The administrator can:

- Add new assets
- Edit asset information
- Delete assets
- View all assets

6.2 User / Department Assignment

The system stores the following assignment details:

- Assigned employee
- Department
- Assignment date

6.3 Maintenance Management

The administrator can:

- Mark a device as under repair
- Update repair status
- Record maintenance date
- Record technician details
- Record maintenance notes

6.4 Search & Filter

Users can search assets by:

- Asset ID
- Serial Number
- Device Type
- Department
- User
- Status

6.5 Reports

The system can generate reports for:

- Operational devices

- Devices under repair
- Maintenance history
- Department-wise assets

7. Non-Functional Requirements

7.1 Performance

- Fast search response
- Support for thousands of records

7.2 Security

- Admin login
- Data validation
- Secure database connection

7.3 Reliability

- Consistent database performance
- Data integrity

7.4 Usability

- User-friendly interface
- Simple navigation

7.5 Availability

- Available during office hours

8. System Features

8.1 Insert Assets

The system stores the following data for each asset:

- Asset ID
- Serial Number
- Device Type
- Brand
- Model
- Purchase Date
- Warranty Expiry

- Assigned User
- Department
- Status

8.2 View Assets

All assets can be displayed with the following capabilities:

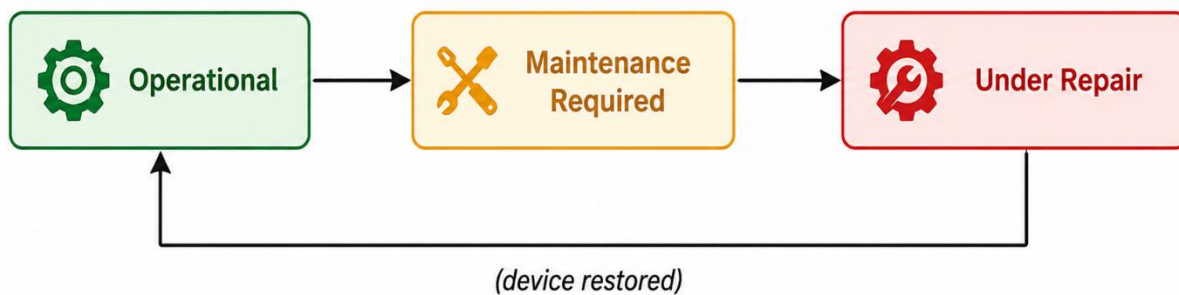
- Search
- Filter
- Sorting

8.3 Update Asset Status

The system supports a defined status lifecycle for every asset, illustrated below:

8.3 Update Asset Status

The system supports a defined status lifecycle for every asset, illustrated below:



8.4 Maintenance Log

Each maintenance record stores:

- Maintenance Date
- Technician Name
- Problem Description
- Repair Details
- Cost
- Completion Date

9. System Modules

Module 1 – Login System

Provides secure administrator authentication before any asset or maintenance data can be accessed or modified.

Module 2 – Dashboard

Displays summary statistics: Total Assets, Operational Devices, Under Repair, and Maintenance Required.

Module 3 – Asset Management

Provides full CRUD (Create, Read, Update, Delete) operations on asset records.

Module 4 – Maintenance Management

Handles maintenance logs and repair history for each asset.

Module 5 – Reports

Generates printable, categorized reports for administrators and department heads.

10. Database Design

The system is supported by a relational database consisting of three core tables, described below.

10.1 Table: Users

Field	Type	Description
UserID	INT (PK)	Unique identifier for each user
FullName	VARCHAR(100)	Full name of the employee
Department	VARCHAR(50)	Department the user belongs to
Email	VARCHAR(100)	Contact email address

10.2 Table: Assets

Field	Type	Description
AssetID	INT (PK)	Unique identifier for each asset
SerialNo	VARCHAR(100)	Manufacturer serial number
DeviceType	VARCHAR(50)	Category of hardware

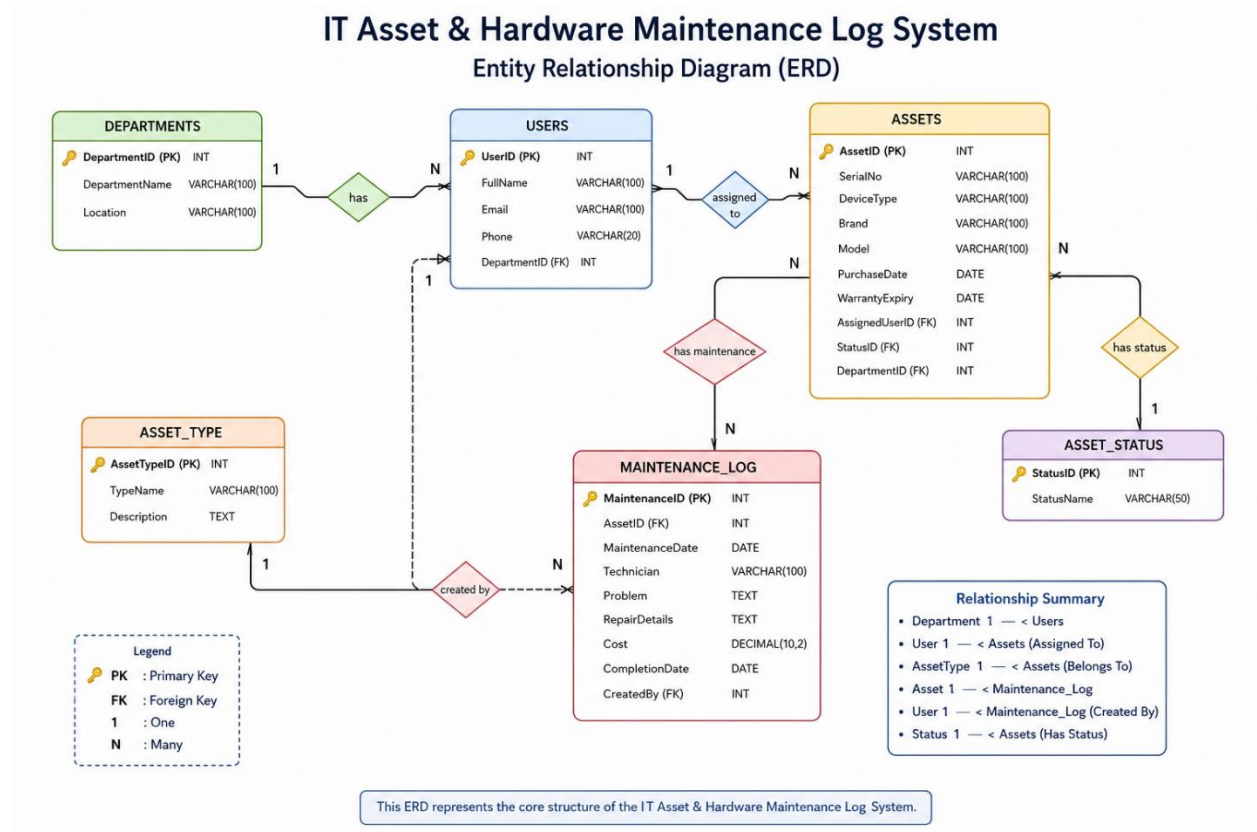
Field	Type	Description
Brand	VARCHAR(50)	Manufacturer / brand name
Model	VARCHAR(50)	Model number or name
PurchaseDate	DATE	Date of purchase
WarrantyExpiry	DATE	Warranty expiration date
AssignedUserID	INT (FK)	References Users.UserID
Status	VARCHAR(30)	Operational / Maintenance Required / Under Repair / Retired

10.3 Table: Maintenance_Log

Field	Type	Description
MaintenanceID	INT (PK)	Unique identifier for each maintenance record
AssetID	INT (FK)	References Assets.AssetID
MaintenanceDate	DATE	Date maintenance was initiated
Technician	VARCHAR(100)	Name of technician performing the work
Problem	TEXT	Description of the reported issue
RepairDetails	TEXT	Description of repair actions taken
Cost	DECIMAL	Cost of the repair
CompletionDate	DATE	Date maintenance was completed

11. Entity Relationship Diagram (ERD)

The diagram below illustrates the relationships between the Users, Assets, and Maintenance Log tables. Each user may be assigned multiple assets and each asset may have multiple maintenance records .

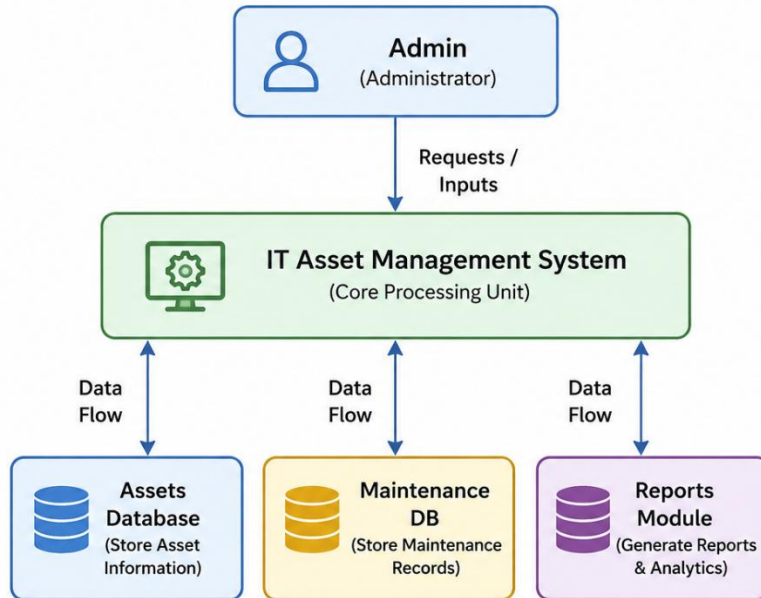


12. Data Flow Diagram (Level 0)

The context-level Data Flow Diagram below shows how the administrator interacts with the system and how data moves between the core processing unit and its databases.

12. Data Flow Diagram (Level 0)

The context-level Data Flow Diagram below shows how the administrator interacts with the system and how data moves between the core processing unit and its databases.



13. Technology Stack

Component	Technology
Frontend	HTML5
Styling	CSS3
Programming (Client-side)	JavaScript
Backend	PHP
Database	MySQL
Web Server	XAMPP
IDE	Visual Studio Code

14. User Interface Overview

14.1 Dashboard

Displays:

- Total Assets
- Operational Devices
- Devices Under Repair
- Maintenance Required
- Recent Maintenance Logs

14.2 Asset Page

Functions:

- Add Asset
- Edit Asset
- Delete Asset
- Search
- Filter

14.3 Maintenance Page

Displays:

- Asset Information
- Maintenance History
- Repair Notes
- Technician Information

14.4 Reports Page

Generates:

- PDF Reports
- Department Reports
- Maintenance Reports

15. Testing Plan

The following test cases will be used to validate core system functionality prior to deployment:

Test Case	Expected Result
Add Asset	Asset saved successfully
Edit Asset	Information updated
Delete Asset	Asset removed
Search Asset	Correct record displayed

Test Case	Expected Result
Update Status	Status changes successfully
Add Maintenance	Maintenance record stored
Generate Report	Report generated correctly

16. Expected Outcomes

After completing this project, the following outcomes are expected:

- All IT assets will be centrally managed.
- Asset assignment becomes transparent.
- Maintenance history is preserved.
- Equipment downtime is reduced.
- Reports are generated quickly.
- Data accuracy is improved.

17. Future Enhancements

Future versions of the system may include:

- QR Code integration
- Barcode scanning
- Email maintenance reminders
- SMS notifications
- Mobile application
- Cloud database
- Multi-user authentication
- Role-based access control
- Dashboard analytics
- Asset depreciation calculation
- Automatic warranty alerts
- REST API integration

18. Conclusion

The IT Asset & Hardware Maintenance Log System is a practical and efficient database management project designed to address the common challenges faced by small IT offices in tracking hardware assets

and maintenance activities. By replacing manual record-keeping with a centralized digital system, the project improves accountability, streamlines maintenance workflows, and enables quick access to accurate asset information.

Its straightforward database structure, essential CRUD operations, and reporting capabilities make it an excellent beginner-to-intermediate DBMS project, while also providing real-world value and a strong addition to a software development or IT support portfolio.

19. References

- Elmasri, R., & Navathe, S. B. Fundamentals of Database Systems.
- Silberschatz, A., Korth, H. F., & Sudarshan, S. Database System Concepts.
- PHP Official Documentation. <https://www.php.net/docs.php>
- MySQL Official Documentation. <https://dev.mysql.com/doc/>
- HTML Living Standard (WHATWG). <https://html.spec.whatwg.org/>
- CSS3 Specification. <https://www.w3.org/Style/CSS/>
- JavaScript (ECMAScript) Documentation. <https://262.ecma-international.org/>